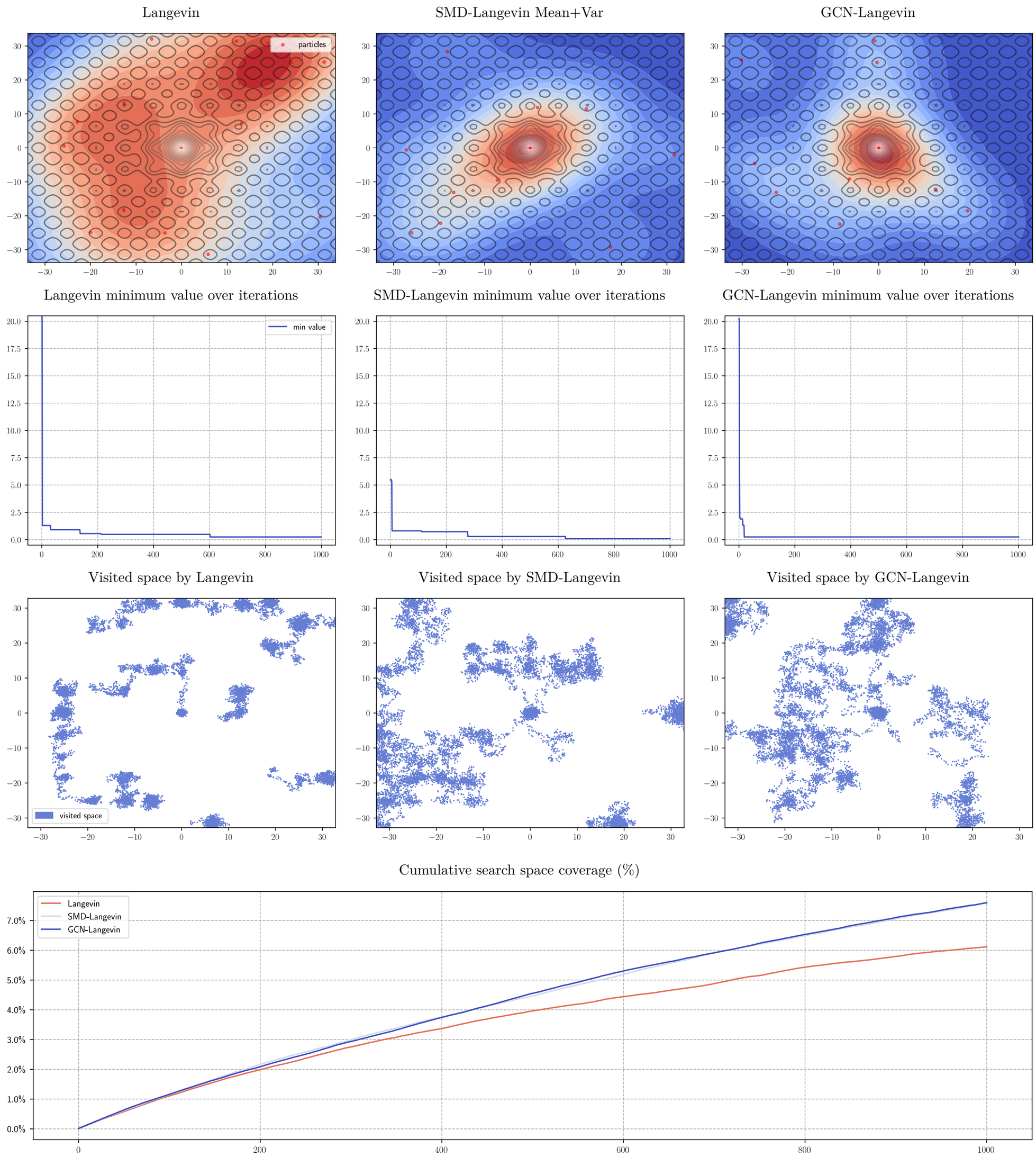

Figure. Visualization of the coverage of the search space for the Langevin, SMD-Langevin and GCN-Langevin algorithms on the Ackley function using 15 particles and 1000 iterations. The first row shows the estimated density (using Gaussian kernels) of particles at the last iteration for each algorithm. The second row shows the evolution of the minimum value of the function over iterations for each algorithm. The third row shows the space that has been visited by each algorithm after 1000 iterations. The last row shows the percentage of the search space that has been visited by the particles over iterations for each algorithm. For the two last rows, the coverage of the search space is computed by discretizing the latter into a ε-cover. We then count the number of balls of radius ε that have been visited by at least one particle during the optimization process. We set $\varepsilon = 0.1$ in our experiments.

One can see that, not only the SMD and GCN variants of Langevin end up with a better concentration of particles around the global minimum, but they also explore more of the search space, as shown by the last row. This suggests that the SMD and GCN variants are able to escape local minima and explore more effectively the search space, leading to better optimization performance.